SPECIFICATION

Super Admin Console

The single place from which staff run the business: set what customers pay, manage every customer end to end, support them, and operate the platform.

COMPANION TO	`voice-agent-portal-spec.md` v0.3 (customer portal). Section IDs there (A*, W*, D*) are referenced here; nothing in this document contradicts it — where this document defines a setting, the portal reads it.
OWNER	Pramod (CTO)
DATE	10 Sep 2026
VERSION	v0.1
AUDIENCE	Metacognixion staff: founders, ops, finance, support; developers building it
ROUTE / HOST	`admin.<domain>` (separate Next.js app in the same monorepo, same backend, `/admin/*` API namespace)

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01 Purpose and principles

The Super Admin console is the single place from which staff run the business: **set what customers pay**, **manage every customer end to end**, **support them**, and **operate the platform** (Retell, Stripe, jobs). Rules:

1. **No direct database edits in production.** Every change customers can feel (price, credits, plan, pause, prompt) is made in this console, requires a **reason**, and is written to the audit log with actor, before/after and timestamp.
2. **One source of truth, real-time sync.** Pricing, rates, caps, templates and feature flags live in versioned config tables. The customer portal never hard-codes them; it reads them through the same API the admin writes. A change published here is visible on the customer side within seconds (cache invalidation) — no deploy.
3. **Least privilege by role.** Support cannot refund $10,000; Finance cannot rewrite a prompt; only a Super Admin can publish pricing or act as a customer.
4. **Everything is explainable to the customer.** Every credit movement done by staff appears in the customer's own ledger (D5) with a plain-English description.
5. **Fast enough to use during a live incident.** Global search, kill switches and pause/resume are one screen away.

02 Staff authentication and roles

2.1 Mechanics

- Separate auth tenant from customers (same provider, different instance/organisation). Sign-in only via **Google Workspace SSO** (company domain) — no passwords. **MFA mandatory** (TOTP or passkey). Session **8 hours**, re-auth for sensitive actions (pricing publish, refunds > $500, act-as impersonation, staff role changes).
- Optional **IP allowlist** (office/VPN) — toggle in SA13.
- Every request carries the staff JWT; backend middleware sets `ctx.staff = {id, role}` and rejects customer JWTs on `/admin/*`.

2.2 Roles and permission matrix

CAPABILITY	SUPER ADMIN	OPS	FINANCE	SUPPORT	READ-ONLY
View all customers, calls, wallets	✓	✓	✓	✓	✓
Listen to / download recordings (reason required)	✓	✓	–	✓	–
Adjust credits	any	≤ 500 / action	any	≤ 50 / ticket, ≤ 200 / day	–
Change plan tier / caps for a tenant	✓	✓	✓	–	–
Pause / resume tenant, number, agent	✓	✓	–	✓ (tenant only, with ticket)	–
Refunds (Stripe)	any	–	≤ $2,000 (> needs SA)	–	–
Custom deals (negotiated price)	✓	–	✓	–	–
Pricing & plans: edit draft	✓	–	✓	–	–
Pricing & plans: **publish**	✓	–	–	–	–
Agent stack (tier→model), templates, guardrails: edit draft	✓	✓	–	–	–
Agent stack: **publish** / migrate all	✓	–	–	–	–
Feature flags, kill switches	✓	✓	–	–	–
View-as customer (read-only)	✓	✓	✓	✓	–
Act-as customer (write, time-boxed)	✓	–	–	–	–
Delete / anonymise customer data	✓	–	–	–	–
Support tickets: reply, assign, close	✓	✓	✓	✓	–
Staff management, IP allowlist	✓	–	–	–	–
Export data / reports	✓	✓	✓	–	–

Limits (e.g. Support ≤ 50 credits per ticket) are themselves settings in SA13 so they can be tuned without a deploy.

03 Navigation and shell



Figure 1 — Admin shell. Global search and the platform health strip sit in the top bar; fourteen sections in the left nav.

- **Global search (⌘K):** matches tenant name/id, user email, phone number (E.164 or partial), Retell call id, Retell agent id, Stripe customer/payment id, ticket number. The result opens the relevant detail page.
- **Health strip:** Retell API reachability + status page, Stripe webhook lag, failed jobs count, concurrency vs allowance. Click → SA12.
- **Impersonation banner** (when active): “Viewing as {tenant} — read-only · Exit” or “ACTING AS {tenant} — 23:10 remaining · reason: … · Exit”.

`/admin`	SA1	Overview
`/admin/customers`	SA2	Customers list · `/admin/customers/[id]/(overview|members|wallet|purchases|agents|numbers|calls|tickets|notes|audit|danger)`
`/admin/users`	SA3	Users (people)
`/admin/support`	SA4	Support inbox · `/admin/support/[ticket]`
`/admin/pricing`	SA5	Pricing & plans (packs, rates, recurring, wallet policy, promo codes, custom deals, Retell cost inputs, versions)
`/admin/agent-stack`	SA6	Tiers → models/voices, templates, guardrails, lint rules, generation prompts
`/admin/telephony`	SA7	Numbers inventory, concurrency, Retell account, KB usage, provisioning queue
`/admin/calls`	SA8	Global calls
`/admin/billing`	SA9	Purchases, refunds, failed top-ups, revenue & COGS reports, tax/MoR
`/admin/comms`	SA10	Announcements, banners, broadcasts, maintenance mode
`/admin/trust`	SA11	Fraud queue, flagged calls, blocklists, KYC (P2), deletion requests
`/admin/system`	SA12	Jobs, webhook events, reconciliation runs, feature flags, kill switches
`/admin/staff`	SA13	Staff users, roles, limits, IP allowlist, settings
`/admin/audit`	SA14	Global audit log

Figure 2 — Sitemap.

04 Screens

SA1 — Overview

Purpose: is the business healthy today?

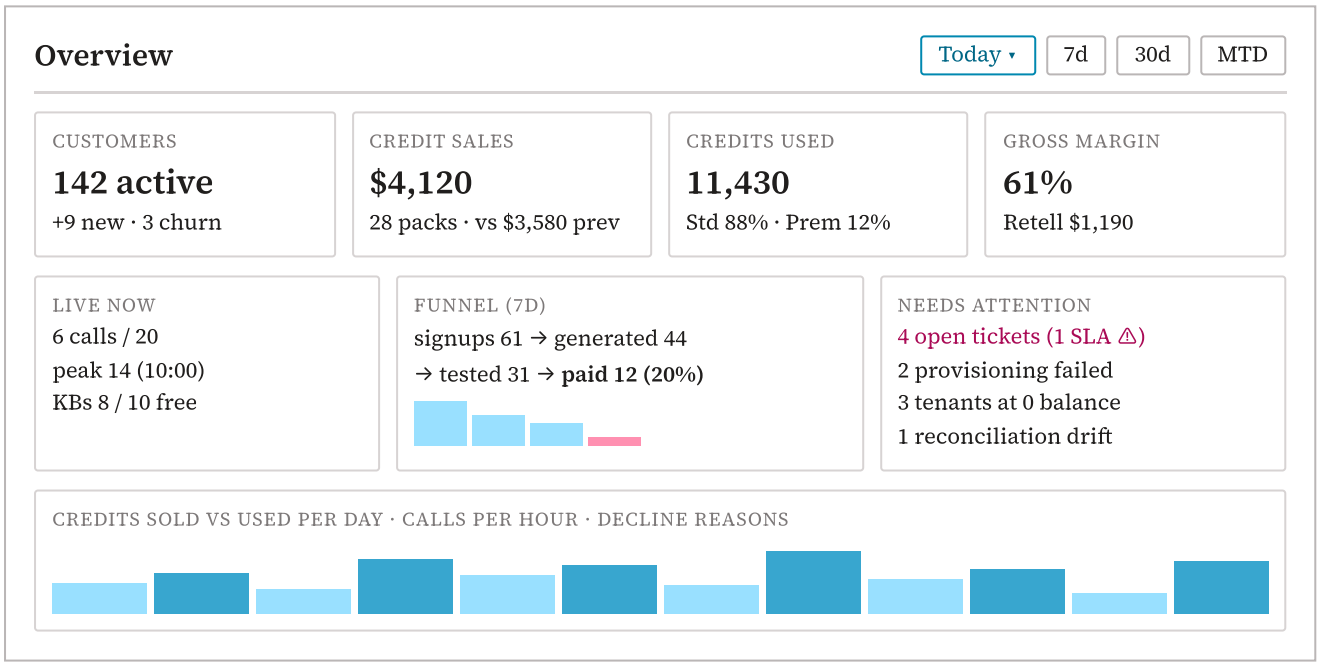


Figure 3 — SA1 Overview. Four business KPIs, live platform state, the onboarding funnel and the "needs attention" queue.

Widgets: customers (active = ≥ 1 call or purchase in 30 d), credit sales ($ and packs, by pack), credits used (Standard vs Premium), gross margin (= credit revenue recognised on usage − Retell `combined_cost` − number/KB/concurrency fixed costs), live concurrency vs allowance, KB count vs free tier, onboarding funnel (W1→W7 events from PostHog or our own step table), "needs attention" list linking into SA4/SA7/SA2/SA12. Definitions in Appendix A. Data: `GET /admin/kpis?range=`.

SA2 — Customers

LIST `/ADMIN/CUSTOMERS`

COLUMN	NOTES
Customer (name, id, country flag)	click → detail
Plan tier · pack history	Trial / Starter / Growth / Scale / Pro / Volume (custom deal badge)
Balance (credits)	red ≤ 0, amber < 20% of last pack
30-d usage (credits) · burn/day · days left	—
Agents (live/draft) · Numbers	—
Status	Active · Onboarding (step n) · Paused (reason) · At risk (0 balance > 3 d) · Churned
Last activity · Owner email	—
Flags	fraud-review · custom-deal · VIP · do-not-contact

Filters: plan, status, country, balance range, signed-up range, last-purchase range, has-open-ticket, flag. Saved views (e.g. "At risk", "Trial > 7 days not activated"). Bulk actions (SA/Ops): send broadcast, add flag, export CSV.

DETAIL `/ADMIN/CUSTOMERS/[ID]` — THE CUSTOMER 360

Header: name, id, plan, balance, status, owner, created, quick actions **[View as] [Adjust credits] [Pause/Resume] [New ticket] [Note]**. Tabs:

TAB	CONTENT	ACTIONS (REASON REQUIRED UNLESS NOTED)
Overview	profile (country, TZ, address, hours, website), lifecycle timeline (signed up → generated → tested → paid → live → …), 30-d stats, alerts sent, open tickets, notes preview, Retell ids	Edit profile (no reason), Change plan tier & caps (concurrency, KB allowance, max agents/numbers — overrides pack defaults), Set flags
Members	users with role, MFA, last login, invites pending	Resend invite, Change role, Remove member, Transfer ownership (SA), Force logout
Wallet	balance, lots (remaining, expiry), full ledger (portal D5 view + staff-only columns: actor, reason), auto top-up config	**Adjust credits** (+/−, reason code, note, optional expiry; shows in the customer ledger as "Credit adjustment — {customer-facing text}"), Extend lot expiry, Toggle/repair auto top-up, Trigger auto top-up now
Purchases	Stripe purchases, invoices, payment methods (masked), failed top-ups	**Refund** (full/partial → Stripe refund + ledger reversal of unconsumed credits; if credits already consumed, refund limited to remaining or requires SA override), Resend receipt, Open in Stripe
Agents	list with tier, languages, voice, published/draft versions, Retell agent/LLM/KB ids, addons	View brief & prompt (any version), View lint, **Force unpublish** (pauses agent — e.g. abuse), Re-run generation (Ops), Change tier (e.g. comp Premium), Toggle addons (denoise/PII), Delete agent (SA)
Numbers	numbers with type, status, rental schedule, Retell number config, inbound webhook health (last 24 h authorisations/declines)	Pause/resume number, Change attached agent, Waive rental (n months), **Release number**, Re-provision (retry), Re-sync from Retell
Calls	portal D2 table + staff columns (Retell cost, decline reason, is_test)	Open call (SA8 detail), **Refund credits for a call** (bad call, reason), Flag call (abuse/quality)
Tickets	tickets for this customer	Open, New ticket
Notes	staff-only notes (markdown, pinned)	Add/edit/pin
Audit	every staff and customer action on this tenant	Export
Danger	Pause tenant (reason; all numbers decline with "paused" reason), Resume, Anonymise & delete (SA; releases numbers, deletes Retell agents/KBs, redacts PII, keeps ledger totals for accounting; 7-day cooling window with cancel)	—

View as / Act as — see §7.4.

SA3 — Users (people)

List of all user accounts across workspaces: email, name, MFA, memberships (workspace + role), last login, status. Actions: Verify email manually, Reset MFA (with identity check note), Lock/unlock account, Merge duplicate (SA), Force logout everywhere. Search by email/name.

SA4 — Support inbox

Full behaviour in §7.

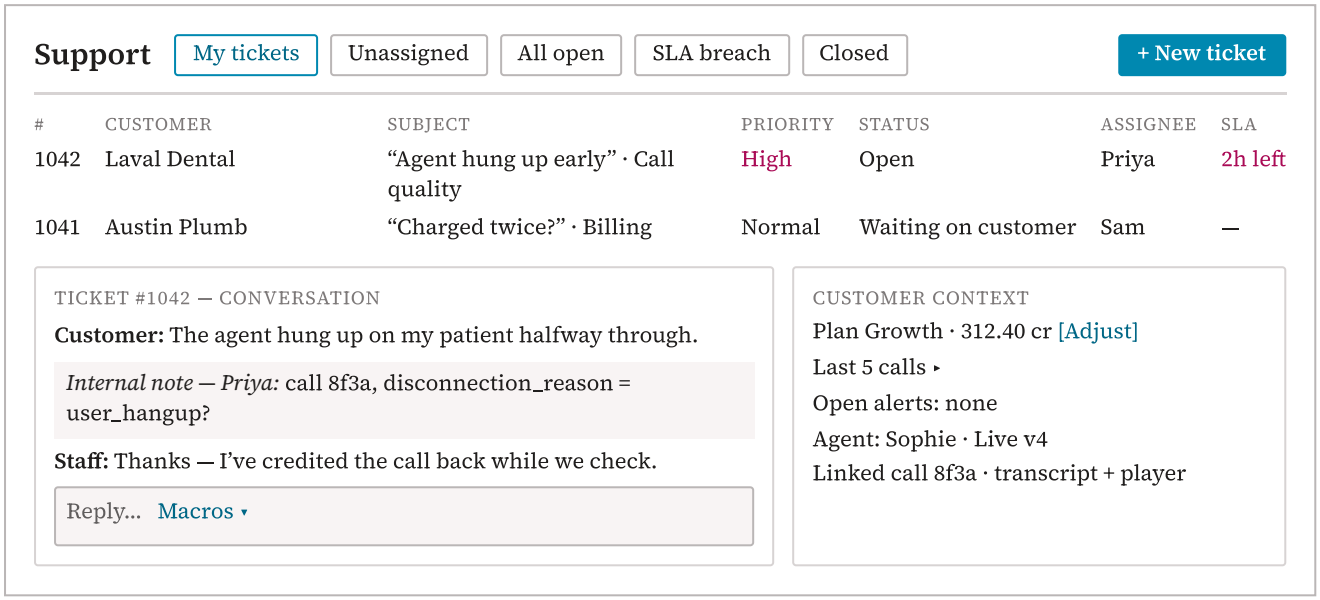


Figure 4 — SA4 Support. Queue views above; the ticket thread and the Customer 360 context panel side by side.

SA5 — Pricing & plans

Sync model in §5.

Sub-tabs: **Packs · Rates · Recurring · Wallet policy · Promo codes · Custom deals · Retell cost inputs · Versions.** Everything edits a **draft**; a diff + margin preview is shown; **Publish** (SA only) creates a new config version.

PACKS (TABLE EDITOR, ONE ROW PER PACK)

FIELD	TYPE	VALIDATION	PORTAL EFFECT
id / display name	text	slug unique	W7 & D5 pack cards
Price · currency	money · USD/CAD	> 0	Stripe Price
Credits (minutes at Standard)	int	> 0	credits granted
Bonus credits (promotional)	int	≥ 0	shown as “+50 bonus”
Plan tier	enum	—	`tenants.plan_tier` after purchase
Concurrency cap · KB allowance · max agents · max numbers	ints	≥ 0	enforced by inbound webhook / portal gates
Feature gates	multi (knowledge_base, premium_voices, multi_agent, pii_removal, denoise, call_my_phone, web_widget, analysis_fields)	—	`GET /flags` per tenant
Badge · description · sort order	text	—	card copy
Visibility	public / hidden / link-only	—	hidden = staff-assignable only
Regions	US, CA, …	—	shown by workspace country
Active	bool	—	inactive packs can’t be bought; existing tiers untouched

RATES

SETTING	DEFAULT	PORTAL EFFECT
Standard tier credits/min	1.00	metering rate (§14 of the portal spec)
Premium tier credits/min	1.50	metering rate; W3 badges
Web-call (browser) credits/min	1.00 (= tier rate)	test & widget calls
Toll-free telephony surcharge credits/min	0.25 (pending verification)	added when the number is toll-free
Transferred-leg credits/min (P2)	0.25	after `transfer_bridged`
Rounding	per second, 2 dp	ledger

Recurring: number rental credits/month (15), extra KB credits/month (30), verified caller-ID one-time credits (P2), release-after-days at zero balance (30), rental warning days (7/14/21).

Wallet policy: trial credits (20), trial validity days (30), trial channel (web only), credit expiry months (12), grace overdraft credits (10), low-balance thresholds (20%, 5%, 0), auto top-up defaults (on/off, threshold 10%), max first purchase without review ($1,000), MFA-required-above balance (1,000).

Promo codes: code, type (% off / fixed off / bonus credits), applies to packs, max uses, one per customer, valid from/to, region, active; usage list. Portal: W7/D5 “Have a promo code?” field.

Custom deals (Volume plan): per tenant — negotiated $/credit or custom pack (price, credits, caps, gates), payment mode (Stripe Checkout / invoice with net-30, credits granted on payment or on approval), contract dates, notes. Creates a tenant-specific hidden pack; appears on the customer’s D5 as their pack.

Retell cost inputs: editable table mirroring portal spec §4 (voice infra, TTS per provider, LLM per model, telephony per country, add-ons, number/KB/concurrency monthly). Used only for **margin previews and COGS estimates** (actuals come from `call_cost`). Update when Retell changes prices.

Margin preview panel: for each pack shows $/credit, Standard and Premium gross margin using the cost inputs and the current tier→model mapping (SA6). Publishing is blocked if any active pack’s Standard margin < a configurable floor (default 35%) unless SA overrides with reason.

Versions: list of published configs (version, published by, at, effective at, change summary); **Compare** any two; **Roll back** (publishes a copy of an older version as new).

SA6 — Agent stack

- **Tiers → providers:** Standard = {LLM model id, fast tier off, TTS providers allowed}, Premium = {…}. Changing the LLM affects **new generations/publishes** (existing published agents keep their Retell config until republished) — or “Migrate all now” (SA, runs a job that PATCHes Retell LLMs, with progress).

- **Templates:** CRUD on the catalogue (portal spec §9.2): brief prefill, steps, analysis fields, rules, default tier/voice per locale, sample audio upload, industries, active. Template versioning; edits affect new agents only.
- **Guardrails text** (portal spec §9.4): edit + version; "Apply to all on next publish" is automatic; "Re-publish all live agents" is an explicit SA action with progress.
- **Lint rules:** thresholds (max prompt chars, greeting words), enable/disable.
- **Generation prompts:** the system prompts used for generate/feedback/greeting-translation, versioned, with a **playground** (paste a brief → see output, no customer impact) and an A/B flag (% of new generations).
- **Voice catalogue:** cached Retell voices with tier override (e.g. mark a specific Cartesia voice Premium), display name, hide/show, default per locale.

SA7 — Telephony & Retell ops

- **Numbers inventory:** every number across tenants; provider, country, status, tenant, agent, rental status, inbound authorisations/declines 24 h, last call. Actions: pause, release, re-sync, move to another tenant (SA, rare).
- **Concurrency:** live gauge (active calls / allowance), hourly peak chart 30 d, per-tenant contribution now; **Buy/release Retell slots —** records target allowance, links to the Retell dashboard (purchase is manual in Retell; the field here keeps our enforcement in sync), alert threshold %.
- **Retell account:** current API key environment, account balance/spend (if the API exposes it; else manual entry + link), status page embed, webhook endpoint health (success rate, p95 latency, retries).
- **Knowledge bases:** count vs free tier, per-tenant KB list, orphaned KBs (no agent) → clean-up.
- **Provisioning queue:** activation/number jobs with step status; Retry, Skip step, Assign to me; failure reason from Retell.
- **Kill switches** (also on SA12): Reject all new inbound (global), Pause web tests, Pause purchases.

SA8 — Calls (global)

Search across tenants by call id, number, tenant, date, decline reason, duration, credits, sentiment, flagged. Detail = portal D2 detail plus: Retell `call_cost` breakdown, our credits charged and rate, agent version and prompt snapshot, tenant, webhook event timeline (started/ended/analyzed received at), raw payload viewer (SA/Ops). Actions: refund credits, flag, add to ticket, **Listen** (requires reason; logged), Delete recording (SA).

SA9 — Billing & finance

- **Purchases:** all Stripe purchases with status, pack, amount, tax, tenant; filters; export. Refund action (limits per role).

- **Failed auto top-ups:** list with Stripe decline code, retry schedule, customer notified?; actions: retry now, disable auto top-up, open ticket.
- **Revenue & COGS reports:** by day/month: credits sold ($), credits consumed (deferred revenue recognised), Retell actual cost (Σ `call_cost`) + fixed (numbers, KBs, slots from SA7), gross margin, by tier, by pack, by country; deferred-credit liability (unconsumed credits × weighted $/credit); expiry breakage. Export CSV/XLSX for accounting.
- **Tax & MoR settings:** merchant entity (Metacognixion CA / Gritbyte IN), Stripe Tax on/off, invoice footer text, currency per region.

SA10 — Communications

- **In-app banners:** message, severity, audience (all / plan / country / list), start/end, dismissible; shows in the portal shell. Preview.
- **Email broadcasts:** template (MJML/markdown), audience segment (same filters as SA2), test send, schedule; unsubscribe respected; sends via the transactional provider (Postmark/SES).
- **Maintenance mode:** portal read-only banner + optional "reject new inbound" kill switch with a customer-facing message.
- **Templates:** edit transactional emails (welcome, receipt, low balance, paused, digest) with variables; version + preview.

SA11 — Trust & compliance

- **Fraud queue:** sign-ups/purchases flagged by rules (disposable email, > $1,000 first purchase, Radar risk, velocity of test calls, many numbers fast). Approve / Block (locks tenant, refunds if paid) / Request info (ticket).
- **Flagged calls:** abuse/quality flags from staff or automated (guardrail violations reported by post-call analysis). Review, action (warn customer via ticket, force unpublish, pause).
- **Blocklists:** email domains, emails, phone numbers (caller blocklist global), countries (inbound) — with reason and expiry.
- **KYC for outbound (P2):** business verification docs, status, reviewer.
- **Data requests:** deletion/export requests (GDPR/PIPEDA/CCPA style), SLA, execute (runs the anonymise job), record.
- **Retention enforcement:** shows tenants' retention settings and the last purge run.

SA12 — System

- **Jobs:** queues (generate, patch, activation, rental, expiry, top-up, reconcile, digest…), counts, failed jobs with stack trace, Retry / Discard, pause queue.
- **Webhook events browser:** Retell/Stripe/auth events, filter by type/tenant/status, payload viewer, **Replay** (re-process idempotently).
- **Reconciliation runs:** nightly results (calls missing/settled, cost drift %), drill-down to affected calls, "Run now".

- **Feature flags:** name, description, default, per-tenant overrides, % rollout; used by the portal `GET /flags`.
- **Kill switches:** global reject inbound, pause web tests, pause purchases, pause generation (LLM outage) — each with confirmation, reason and auto-expiry option.
- **Environment info:** versions deployed, Retell/Stripe/Anthropic key environments, last migration.

SA13 — Staff & settings

Staff users (email, role, MFA, last login), invite via Google SSO domain, change role, deactivate. Role limits (credit adjustment caps, refund caps, ticket SLAs). IP allowlist. Support settings (support email, signature, business hours for the SLA clock, canned responses library). Notification routing (Slack channels for failed jobs, fraud, SLA breach).

SA14 — Audit log (global)

Every staff action and every sensitive customer action: time, actor (staff or customer user), tenant, action, target, reason code + note, before/after JSON, IP, impersonation context. Filters + export. Immutable (append-only table; no UPDATE/DELETE grants).

05 Pricing, plans and rates — how configuration syncs end to end

5.1 Config model

- `pricing_configs` holds **versioned, immutable** JSON snapshots: packs, rates, recurring, wallet policy, feature-gate defaults per tier. Exactly one version is `current` (highest `effective_at ≤ now`). Drafts live in `pricing_drafts` until published.
- `promo_codes`, `custom_deals`, `feature_flags`, `templates`, `agent_stack_configs`, `guardrail_versions` are separate versioned tables with the same draft → publish pattern where customer-visible.

5.2 Publish sequence (SA5 → everywhere)

1. Staff edits draft → **Preview diff + margin check** → **Publish** (SA, re-auth).
2. Backend transaction: insert `pricing_configs` (version n+1, `effective_at` = now or scheduled) + `audit_log`.
3. **Stripe sync job:** for each pack with a changed price/currency → create a new Stripe Price (archive the old); packs are Stripe Products keyed by pack id. Custom deals create tenant-specific Prices. Checkout sessions always use the Price id stored in the current config, so an in-flight session keeps its price.
4. **Cache invalidation:** Redis keys `pricing:current`, `flags:*`, per-tenant `caps:{ws}` are dropped; portal `GET /pricing` and `GET /flags` re-read the config (public CDN cache max-age 60 s with purge).
5. **Enforcement sync:** the inbound webhook reads caps from `caps:{ws}` (tier defaults from the current config, overridden by tenant-level overrides from SA2). Rates are read at `call_ended` time from the config **that was current when the call started** (stored on the `calls` row as `rate` and `config_version`) so a mid-call price change never double-bills.
6. **Grandfathering rules** (chosen at publish time): "Existing tenants keep their current caps until next purchase" (default) or "Apply new tier caps to all tenants now". Credits already purchased are never re-priced.
7. **Customer-facing change log** (optional toggle): publishes a banner "Our pricing changed on {date}" via SA10.

Figure 5 — Publish sequence. Step 3 (magenta) is the only step that leaves our systems; everything else is config and cache.

5.3 What reads what

PORTAL SURFACE	READS	REFRESH
W7 / D5 pack cards, prices, badges, "1 credit = …" copy	`GET /pricing` (current config, region-filtered)	≤ 60 s
W3 voice tier badges and credit rates	rates + voice catalogue tier	≤ 60 s
Plan gates (KB, multi-agent, premium voices, addons)	`GET /flags` (tier defaults ∘ tenant overrides ∘ feature flags)	immediate on tenant page load
Inbound webhook caps and grace	`caps:{ws}`	immediate
Metering rate	`calls.rate` captured at `call_started` from the current config	per call
Rental/KB recurring charges	recurring settings at job run time	next run
Trial credits, expiry months	wallet policy at grant time (stored on the lot)	per grant
Templates, guardrails, tier→model	agent-stack config at generate/publish time (version stored on `agent_versions`)	per generation/publish

5.4 Guardrails on pricing changes

- Margin floor check (SA5) using the Retell cost inputs; block or SA override with reason.
- Cannot delete a pack that any tenant bought in the last 12 months — deactivate instead.

- Cannot reduce a tenant's cap below their current live usage without pausing (warns).
- All publishes emit a Slack notification with the diff.

06 Customer management — end to end

Lifecycle and where staff act:

STAGE	SIGNALS IN ADMIN	STAFF TOOLS
Sign-up / trial	SA1 funnel; SA2 "Trial > 7 d not activated" view; fraud queue (SA11)	Approve/block, extend trial credits (adjust), nudge broadcast (SA10), open ticket
Onboarding	tenant `onboarding_step`; generation failures (SA12 jobs)	View-as to see what they see, re-run generation, edit brief on their behalf (act-as), fix voice/language
Activation	activation job steps (SA7 queue); Stripe payment (SA9)	Retry provisioning, waive rental, change area code, manual number assignment
Live	calls, decline reasons, sentiment, alerts (SA2 Overview)	Adjust credits, change caps, comp Premium, force unpublish, pause number, listen to flagged calls
Billing events	failed top-ups, refund requests, disputes (Stripe webhook `charge.dispute.created` → ticket auto-created)	Refund, retry top-up, custom deal, invoice mode
At risk / churn	0 balance > 3 d, no calls 14 d, number release countdown	Outreach ticket, promo code, extend expiry, pause instead of release
Offboarding	deletion request (SA11)	Anonymise & delete with cooling window; export data

Every action: reason code (Appendix B) + free-text note → audit log → customer-visible ledger text where money or credits are involved.

Customer-visible sync: credit adjustments appear instantly in the D5 ledger and the balance pill; plan/caps changes update the D1 plan label; pause shows the D1 banner "Your account is paused — contact support" with the ticket link; tickets appear in the customer's Help page (D7 ▸ My tickets).

07 Support — ticketing, macros, impersonation

7.1 Ticket sources

- Customer portal: D7 ▸ **Contact support** form (category, subject, message, attachments) and **Report a problem** on a call (D2) → ticket pre-linked to the call.
- Email to support@ (inbound parsing via the transactional provider) → ticket; replies thread by ticket id in the subject.
- System-generated: failed auto top-up ×3, provisioning failed, dispute, fraud "request info", SLA-breaching alerts.
- Staff-created (SA2 ▸ New ticket, e.g. outreach).
- (Optional P2) Live chat via Intercom/Crisp with a webhook that mirrors conversations into tickets and a sidebar app showing the Customer 360.

7.2 Ticket model

Fields: number, tenant, requester (user), subject, category (Billing / Call quality / Setup help / Phone number / Account & login / Feature request / Abuse / Other), priority (Low/Normal/High/Urgent), status (New / Open / Waiting on customer / Waiting on us / Resolved / Closed), assignee, SLA (first response: Urgent 1 h, High 4 h, Normal 1 business day; resolution targets configurable), linked call ids, linked purchase ids, tags, satisfaction rating (after close).

7.3 Ticket detail — layout and actions

- **Thread:** customer messages, staff replies (emailed to the requester + visible in D7), **internal notes** (staff only), system events (macro run, status change).
- **Reply box:** rich text, attachments, canned responses (search), **Macros** dropdown, Send & set status.
- **Customer context panel:** plan, balance (with **Adjust credits** inline, limited per role), last 5 calls (open transcript/player in a drawer), open alerts, agent status (live/draft/paused), number status, recent purchases, previous tickets, notes.

MACROS (ONE CLICK, EACH AUDITED AND EACH POSTS A SYSTEM EVENT IN THE THREAD)

MACRO	WHAT IT DOES
Comp credits (n)	ledger + `adjustment`, reason `goodwill`, customer-facing text
Refund credits for linked call	reverses the call's debit
Resend welcome / receipt / forwarding guide email	transactional resend
Re-run activation / re-provision number	enqueues the job, links the result
Force republish agent	publish latest draft (with customer consent noted)
Pause / resume tenant or number	with reason
Extend credit expiry (+30/60/90 d)	on selected lots
Send forwarding instructions for carrier X	inserts the guide into the reply
Escalate to Ops / Finance / SA	reassign + priority + Slack ping

7.4 Impersonation

- **View as (read-only):** opens the customer portal in a new tab with a staff session flagged `impersonate=view`, banner shown, all write endpoints return 403. Available to SA/Ops/Finance/Support. Reason optional, logged.
- **Act as (write):** SA only; requires reason code, ticket link, re-auth; **30-minute** window (extend once); banner with countdown; every write is audited with `impersonated_by`; the customer Owner receives an email "A support staff member made changes to your account (ticket #…)" — toggle in SA13.
- Never available for: purchases with the customer's card, changing the customer's password/MFA, deleting the workspace.

7.5 SLA and routing

- Auto-assignment round-robin within the Support role; Urgent pings Slack; the SLA clock respects support business hours (SA13). Breach → SA1 "needs attention" and Slack.
- Customer satisfaction: 1-click rating email on close; shown on SA1 (CSAT 30 d).

08 Data model additions

Additions to portal spec §15.

TABLE	KEY FIELDS
staff_users	id, auth_provider_id, email, name, role (super_admin/ops/finance/support/readonly), mfa_enforced, active, last_login_at
staff_role_limits	role, credit_adjust_max_per_action, credit_adjust_max_per_day, refund_max_cents, can_listen_recordings, …
pricing_drafts	id, based_on_version, config_json, edited_by, updated_at
pricing_configs	version, config_json (packs, rates, recurring, wallet_policy, tier_defaults), published_by, published_at, effective_at, change_summary, grandfather_mode
stripe_price_map	pack_id, currency, config_version, stripe_product_id, stripe_price_id, active
promo_codes	id, code, type, value, applies_to_packs[], max_uses, per_customer_once, valid_from, valid_to, regions[], active, uses_count
custom_deals	id, tenant_id, name, price_cents, currency, credits, caps_json, gates_json, payment_mode (checkout/invoice), stripe_price_id, contract_start, contract_end, notes, created_by
tenant_overrides	tenant_id, concurrency_cap, kb_allowance, max_agents, max_numbers, gates_json, flags_json, reason, set_by, updated_at
agent_stack_configs	version, tiers_json (model, tts providers), voice_overrides_json, guardrails_text, lint_json, generation_prompts_json, published_by, published_at
templates	(portal §15) + version, active, updated_by
retell_cost_inputs	version, costs_json, updated_by, updated_at
credit_adjustments	id, tenant_id, ledger_id, delta, reason_code, note, customer_text, staff_id, ticket_id, impersonation_id
refunds	id, purchase_id, amount_cents, credits_reversed, stripe_refund_id, reason_code, staff_id, status
support_tickets	id (number), tenant_id, requester_user_id, subject, category, priority, status, assignee_staff_id, sla_first_response_due, sla_resolution_due, first_responded_at, resolved_at, closed_at, csat, tags[], linked_call_ids[], linked_purchase_ids[]
ticket_messages	id, ticket_id, author_type (customer/staff/system), author_id, body, is_internal, attachments_json, created_at
canned_responses	id, title, body, category, updated_by
customer_notes	id, tenant_id, staff_id, body, pinned, created_at

TABLE	KEY FIELDS
`customer_flags`	tenant_id, flag (fraud_review/custom_deal/vip/do_not_contact/...), set_by, note
`impersonation_sessions`	id, staff_id, tenant_id, mode (view/act), reason_code, ticket_id, started_at, expires_at, ended_at
`announcements`	id, type (banner/email), body, severity, audience_json, starts_at, ends_at, dismissible, created_by
`email_templates`	key, subject, body, version, updated_by
`feature_flags`	key, description, default_on, rollout_pct, tenant_overrides_json
`kill_switches`	key, on, reason, set_by, expires_at
`blocklists`	type (email/domain/phone/country), value, reason, expires_at, set_by
`fraud_reviews`	id, tenant_id, trigger, score, status, reviewer_id, decision, note
`data_requests`	id, tenant_id, type (delete/export), requested_by, due_at, executed_at, job_id
`staff_audit_log`	id, staff_id, action, target_type, target_id, tenant_id, reason_code, note, before_json, after_json, ip, impersonation_id, created_at (append-only)

Portal-side additions implied: `calls.rate`, `calls.config_version`, `credit_lots.policy_version`, `agent_versions.stack_version`, `tenants.paused_reason`, `tenants.custom_deal_id`.

09 Admin API

Namespace `/admin`, staff JWT.

AREA	ENDPOINTS
KPIs	`GET /admin/kpis?range=` · `GET /admin/kpis/funnel`
Customers	`GET /admin/customers` · `GET /admin/customers/:id` · `PATCH /admin/customers/:id` (profile, flags) · `PUT /admin/customers/:id/overrides` · `POST /admin/customers/:id/pause` · `POST …/resume` · `POST …/credits/adjust` · `POST …/lots/:lotId/extend` · `POST …/anonymise` · `POST …/impersonate` {mode, reason, ticket_id} · `DELETE /admin/impersonations/:id`
Members /users	`GET /admin/users` · `POST /admin/users/:id/(verify-email|reset-mfa|lock|unlock|force-logout)` · `PATCH /admin/customers/:id/members/:mid`
Wallet/billing	`GET /admin/purchases` · `POST /admin/purchases/:id/refund` · `GET /admin/topups/failed` · `POST /admin/topups/:id/retry` · `GET /admin/reports/(revenue|cogs|deferred|breakage)` · `PUT /admin/billing/settings`
Pricing	`GET /admin/pricing/draft` · `PUT /admin/pricing/draft` · `GET /admin/pricing/preview` (diff + margins) · `POST /admin/pricing/publish` {effective_at, grandfather_mode} · `GET /admin/pricing/versions` · `POST /admin/pricing/rollback/:version` · `CRUD /admin/promo-codes` · `CRUD /admin/custom-deals` · `PUT /admin/retell-costs`
Agent stack	`GET/PUT /admin/agent-stack/draft` · `POST /admin/agent-stack/publish` · `POST /admin/agent-stack/migrate-all` · `CRUD /admin/templates` · `POST /admin/generation/playground` · `PUT /admin/voices/:id`
Telephony	`GET /admin/numbers` · `POST /admin/numbers/:id/(pause|resume|release|resync|reprovision)` · `GET /admin/concurrency` · `PUT /admin/concurrency/allowance` · `GET /admin/kbs` · `GET /admin/provisioning` · `POST /admin/provisioning/:jobId/retry`
Calls	`GET /admin/calls` · `GET /admin/calls/:id` · `POST /admin/calls/:id/(refund|flag)` · `GET /admin/calls/:id/recording` {reason}
Support	`GET /admin/tickets` · `POST /admin/tickets` · `GET /admin/tickets/:id` · `POST /admin/tickets/:id/messages` · `PATCH /admin/tickets/:id` (status, assignee, priority, tags) · `POST /admin/tickets/:id/macros/:macro` · `CRUD /admin/canned-responses` · customer side: `POST /support/tickets`, `GET /support/tickets`, `POST /support/tickets/:id/messages`
Comms	`CRUD /admin/announcements` · `POST /admin/broadcasts` · `PUT /admin/maintenance` · `CRUD /admin/email-templates`
Trust	`GET /admin/fraud` · `POST /admin/fraud/:id/(approve|block|request-info)` · `CRUD /admin/blocklists` · `GET /admin/flagged-calls` · `CRUD /admin/data-requests`
System	`GET /admin/jobs` · `POST /admin/jobs/:id/(retry|discard)` · `GET /admin/webhook-events` · `POST /admin/webhook-events/:id/replay` · `GET /admin/reconciliations` · `POST /admin/reconciliations/run` · `CRUD /admin/flags` · `PUT /admin/kill-switches/:key`
Staff	`GET/POST/PATCH /admin/staff` · `PUT /admin/staff/limits` · `PUT /admin/settings`
Audit	`GET /admin/audit` · `GET /admin/audit/export`

Portal endpoints that read admin-owned config (public/tenant JWT): `GET /pricing`, `GET /flags`, `GET /announcements/active`, `POST /wallet/checkout` {promo_code?}, `GET /support/tickets`.

10 Security and audit

- Separate host and auth organisation; SSO + MFA only; 8-h sessions; re-auth for sensitive actions; optional IP allowlist.
- Role limits enforced server-side (never only in the UI); exceeding a limit returns 403 with "requires {role}" and offers **Request approval** (creates an approval task for SA/Finance — P2).
- **Reason required** on every customer-affecting write; reason codes standardised (Appendix B).
- Recording access is logged with staff id, reason and ticket; recordings stream via short-lived signed URLs; Support sees PII-masked transcripts by default (numbers/emails partially masked) with "Reveal" logged.
- Impersonation rules §7.4; no card actions; customer notification on act-as.
- Audit log append-only; retained 7 years; exportable; tamper-evident (hash chain per day — P2).
- Admin app behind a WAF; rate limits; CSP; no customer data in URLs; exports watermarked with the staff id.

11 Phasing

P1

with portal MVP

SCOPE	SA1 (core KPIs), SA2 list + 360 (Overview, Members, Wallet adjust, Purchases refund, Agents view/force-unpublish, Numbers pause/release/retry, Calls, Notes, Audit, Pause/Resume), SA3 basics, SA4 built-in tickets + macros (comp credits, refund call, resend emails, re-run activation), SA5 Packs/Rates/Recurring/Wallet policy with draft→publish + Stripe sync + margin preview, SA7 numbers/concurrency/provisioning queue, SA8, SA9 purchases/refunds/failed top-ups/basic revenue report, SA12 jobs/webhook replay/reconciliation/kill switches/flags, SA13, SA14, view-as impersonation
WHY	Needed to run 10 design partners without touching the DB

P2

SCOPE	Promo codes, custom deals, SA6 full (templates editor, playground, migrate-all), SA10 banners/broadcasts/email templates, SA11 fraud queue/blocklists/KYC, act-as impersonation, SLA automation + CSAT, Intercom mirror, approval workflow for over-limit actions, revenue/COGS/deferred reports
WHY	Growth and finance rigour

P3

SCOPE	Agency/reseller sub-accounts, API-key management for the public API, multi-region pricing, tamper-evident audit, SOC 2 evidence exports
WHY	Scale

A KPI definitions

KPI	DEFINITION
Active customers	tenants with ≥ 1 connected call or purchase in trailing 30 d
Credit sales	Σ paid purchases (excl. refunds) in range, in reporting currency
Credits used	Σ call debits (excl. test calls flag toggle) in range
Recognised revenue	credits used × weighted $/credit of the lots consumed (FIFO)
COGS	Σ Retell `call_cost.combined_cost` for calls in range + prorated fixed costs (numbers $2, KBs $8 beyond 10, slots $8)
Gross margin	(Recognised revenue − COGS) ÷ Recognised revenue
Deferred liability	Σ unexpired `credit_lots.credits_remaining` × their $/credit
Breakage	credits expired in range × $/credit
Activation rate	tenants that paid ÷ tenants that signed up (cohort by sign-up week)
Time to live	median minutes sign-up → `onboarding_completed_at`
Decline rate	declined inbound ÷ (declined + connected), by reason
CSAT	avg rating on closed tickets, trailing 30 d

B Reason codes

For adjustments, refunds, pauses and access.

`goodwill · service_issue (bad call / outage) · billing_error · promo · partner_deal · fraud_suspected · abuse_policy · customer_request · chargeback · expiry_extension · testing_internal · support_investigation (recording access) · legal_request · other (note mandatory)`

C Cross-reference and required portal-spec changes

PORTAL SPEC ITEM	CHANGE TO KEEP BOTH DOCUMENTS IN SYNC
D8 Internal admin console	Replaced by this document (pointer added in portal spec v0.3)
D7 Help	Add **Contact support** form and **My tickets** list (§7.1)
D2 Calls detail	“Report a problem” creates a support ticket linked to the call (§7.1)
W7 / D5 Buy credits	Add optional **Promo code** field (SA5); price/pack data comes from `GET /pricing`
D1 Home	Paused banner text and support ticket link when the tenant is paused by staff (§6)
§5 Business model	All numeric values (pack prices, credits, rates, rental, trial, expiry, grace, thresholds) are **defaults of the pricing config**, editable in SA5 without deploy
§9 Prompt system	Templates, guardrails, lint thresholds, tier→model mapping are admin-managed (SA6); `agent_versions.stack_version` records which version generated them
§14 Metering	`calls.rate` and `calls.config_version` captured at `call_started`; caps read from `caps:{ws}` = tier defaults ∘ tenant overrides
§15 Data model	Add the tables in §8 and the portal-side columns listed there
§16 API	Add `GET /pricing`, `GET /announcements/active`, `POST /support/tickets`, `GET /support/tickets`, promo code on checkout
§17 Security	Staff access to customer data governed by §10 (reason-logged recording access, masked PII for Support)